

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I M.Kyathi Sai Priya (192572101) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ___M.Kyathi_

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1.Explain the difference between HDFS and traditional file systems.

Answer:

HDFS is designed for large-scale distributed data storage, while traditional file systems such as NTFS or ext4 usually store data on a single machine or storage system.

- HDFS divides large files into blocks and distributes them across multiple nodes.
- It provides automatic data replication and fault tolerance.
- Traditional file systems generally depend on the reliability of one machine/storage device.
- HDFS is optimized for high-throughput, large sequential reads/writes, rather than low-latency access to small files.

Architecture: HDFS uses a NameNode to manage metadata and DataNodes to store actual data blocks.

2.Why is replication important in distributed storage systems?

Answer:

Replication creates multiple copies of the same data on different nodes. If one node fails, another copy can still be accessed.

For example, with a replication factor of 3, HDFS stores three copies of a block on different DataNodes.

Benefits:

- Fault tolerance
- High availability
- Data recovery
- Improved read performance

3.How does MapReduce divide work between map and reduce phases?

Answer:

MapReduce processes large datasets in two main stages:

Map phase:

The input data is divided into smaller pieces, and mapper tasks process them in parallel to generate key-value pairs.

Shuffle and Sort:

The intermediate results are grouped according to their keys.

Reduce phase:

Reducers process the grouped data and generate the final output.

Example: For counting words, Mapper produces (word, 1), and Reducer adds all values for the same word.

4. What role does YARN play in large-scale Hadoop processing?

Answer:

YARN (Yet Another Resource Negotiator) manages the computing resources and applications in a Hadoop cluster.

- Allocates CPU and memory resources.
- Schedules applications.
- Monitors running tasks.
- Allows different processing frameworks to run on the same Hadoop cluster.

Architecture: The ResourceManager manages cluster resources, while NodeManagers manage resources on individual nodes.

5. Differentiate NAS and SAN in an enterprise data environment

Feature	NAS	SAN
Access	File-level	Block-level
Connection	Usually Ethernet/IP	Usually Fibre Channel/iSCSI
Management	Easier	More complex
Typical use	File sharing, backups	Databases, enterprise applications
User view	Shared files/folders	Appears as storage disks

Simple explanation: NAS provides shared files, whereas SAN provides shared block storage.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL means Extract, Transform, Load.

- Extract: Collect data from different sources.
- Transform: Clean, validate, filter, and convert the data.
- Load: Store the processed data in a target system such as a data warehouse or data lake.

Purpose: ETL converts raw data into clean, consistent, analysis-ready information.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data-flow and integration tool used to move and process data between different systems.

It provides:

- Visual flow-based design
- Data routing and transformation
- Integration with databases, APIs, files, and Hadoop systems
- Monitoring and tracking of data movement
- Back-pressure and error handling

Architecture: NiFi acts as a controlled pipeline between data sources → processing → storage/analytics systems.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

Common failure points include:

- DataNode failure: HDFS automatically recreates missing replicas.
- NameNode failure: Use NameNode high availability with an active and standby NameNode.
- Network failure: Use redundant network paths and monitor connectivity.
- Disk failure: Detect failed disks and replace them while relying on replicated data.
- Resource failure: Monitor CPU, memory, and disk utilization.

The main strategy is replication + monitoring + automatic recovery + redundancy.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would design it in multiple security layers:

Data Sources → Secure Ingestion → Validation/Transformation → Encrypted Storage → Processing → Access/Reporting

Key controls:

- Use TLS encryption during data transmission.
- Encrypt sensitive data at rest.
- Use authentication and role-based authorization.
- Validate and sanitize incoming data.
- Maintain audit logs.
- Apply data-access policies and least-privilege permissions.
- Monitor the pipeline for suspicious activity.

Architecture principle: Security should be implemented at every layer, rather than only at the storage layer.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

“I would explain Hadoop as a system that allows us to store and process very large amounts of data using many computers working together.

Instead of buying one extremely powerful computer, we can add more standard computers as our data grows. Hadoop automatically distributes the data and workload across these machines.

If one machine fails, replicated data allows the system to continue working. Therefore, the architecture provides scalability, fault tolerance, and cost-effective large-scale processing.”

